

Lab 1: [Basic Logic Design]

Requirements Not Met

N/A

Problems Encountered

I found it difficult to keep track of the truth table during the pre-lab.

Applications

We learned how to translate verbal proclamations into actual circuits using logic gates.

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Pre-Lab Questions

$$2. F = (A \cdot B) + (A \cdot \bar{B}) + (\bar{A} \cdot B)$$

$$AB + A\bar{B} \quad \text{Factor out}$$

$$A(B + \bar{B}) \quad \text{Inverse law}$$

$$A \cdot 1 = A \quad \text{Identity law}$$

$$A + (\bar{A} \cdot B) \quad \text{Distributive law}$$

$$(A + \bar{A})(A + B) \quad \text{Inverse law}$$

$$(1)(A + B) \quad \text{Identity law}$$

$$\underline{A + B} \quad \text{MSOP}$$

$$6. \bar{V}W + X + \bar{Y}(X + \bar{V}W) \quad \text{Distributive law}$$

$$\bar{V}W + X + \bar{Y}X + \bar{Y}(\bar{V}W) \quad \text{Absorption}$$

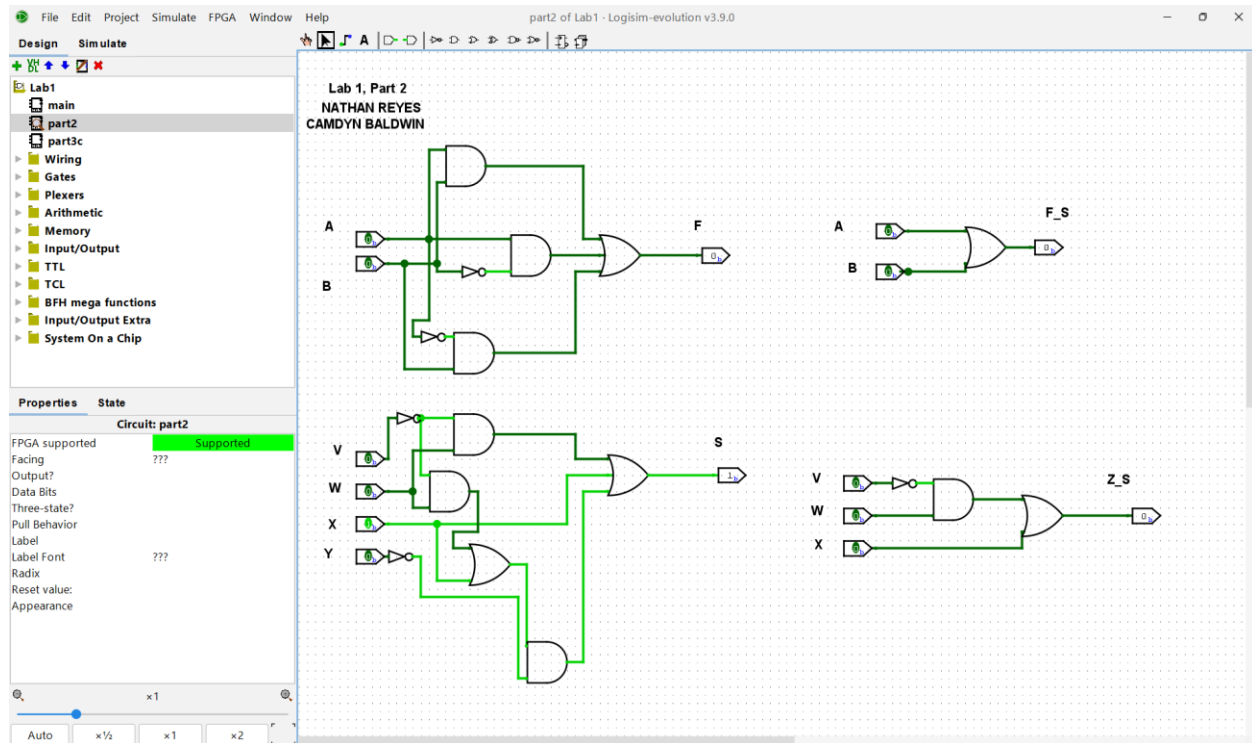
$$\bar{V}W + \bar{Y}(\bar{V}W) = \bar{V}W$$

$$\bar{V}W + X + \bar{Y}X \quad \text{Absorption}$$

$$X + \bar{Y}X = X$$

$$\underline{\bar{V}W + X} \quad \text{MSOP}$$

Part 2.



Part 3.

a.

D	W	M	A	S
T	T	T	F	T
F	T	T	T	T
F	F	T	T	T
F	F	F	F	T
T	F	F	F	T
T	T	F	F	F
T	F	T	T	T
F	T	F	F	T

b. $A = M \cdot (\bar{D} + \bar{W})$
 $S = M + (\bar{D} + \bar{W})$

$$C. A(D, W, M) =$$

$$(\bar{D}WM) + (\bar{D}\bar{W}M) + (D\bar{W}M) \quad \text{Distributive}$$

$$M(\bar{D}W + \bar{D}\bar{W} + D\bar{W}) \quad \text{Distributive}$$

$$M(\bar{D}W + (\bar{D} + D)(\bar{W})) \quad \text{Inverse}$$

$$M(\bar{D}W + \bar{W}) \quad \text{Distributive}$$

$$M(\bar{D} + \bar{W})(W + \bar{W}) \quad \text{Inverse}$$

$$\underline{M \cdot (\bar{D} + \bar{W})} \quad \text{SOP}$$

$$S(D, W, M) =$$

$$(\bar{D} + \bar{W} + M) \quad \text{Commutative}$$

$$\underline{M + \bar{D} + \bar{W}} \quad \text{SOP}$$

